

Natural Disaster Severity Prediction via Leakage-Aware Boosted Trees and Temporal Feature Engineering

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Abstract—Group ID: 5. Members: Shin-Yu Chen (314551141, sychen.cs14@nycu.edu.tw), Velda Wei-Hsin Hung (314553021, velda.cs14@nycu.edu.tw), and Sky Shih-Kai Hong (314554024, sky.cs14@nycu.edu.tw). Our code is available at GitHub: <https://github.com/skyhong2002/disaster-severity-prediction>.

This project predicts weekly drought severity scores for 2,248 regions from historical meteorological observations. The central difficulty is not only multi-horizon forecasting, but also preventing target leakage from weekly score labels while adapting to train-test season and weather shift. We implement direct horizon models using LightGBM, XGBoost, and CatBoost, with calendar, lag, rolling, exponentially weighted, drought-proxy, climatology, and 91-day-gapped score-history features. The best reproducible reportable submission is a 35/35/30 LightGBM/XGBoost/CatBoost ensemble with Kaggle public MAE 0.8124. A separate already-submitted public-chase horizon hedge reaches public MAE 0.7905 and crosses Baseline 3, but we keep it separate from the method claim because it was selected by public leaderboard feedback rather than by a new training algorithm.

I. PROJECT SUMMARY (MAX 1 PAGE)

A. Goal

The competition task is to predict the drought severity score for each region over the next five consecutive weeks after the 91-day test observation window. Scores are on a 0–5 scale, with larger values indicating more severe drought. The submission must contain one row per `region_id` and five numeric columns, `pred_week1` through `pred_week5`. The evaluation metric is mean absolute error (MAE).

B. Approach

We frame the task as supervised direct multi-step regression. For each horizon $h \in \{1, \dots, 5\}$, the training label is the real weekly score h weeks after a forecast origin. The final pipeline builds a time-safe feature table from raw weather, recent and long-window weather summaries, calendar variables, region-month climatology anomalies, and score-history features shifted at least 91 days. The final reportable model is a convex ensemble of three boosted-tree families: LightGBM, XGBoost, and CatBoost. We chose this design because tabular gradient boosting remains highly competitive on heterogeneous structured features, while being easier to audit than large neural sequence models.

C. Related Work and SOTA Context

Our method is closest to the competition-style tabular forecasting tradition: XGBoost [1], LightGBM [2], and CatBoost [3] are strong baselines for mixed continuous/categorical data and sparse feature interactions. The most relevant sequence alternatives for this assignment are TCN-style convolutional models [4], PatchTST Transformers [5], and the simple DLinear baseline [6], because peer reports explicitly compared neural sequence models with tree or linear baselines. Recent foundation models such as TimesFM [7] show the broader SOTA direction, but our dataset has a specific supervised target, high-cardinality regions, and strict leakage constraints, so zero-shot generality is less central than validation reliability. In drought forecasting, prior reviews emphasize that meteorological drought is nonlinear, nonstationary, and often benefits from hybrid or ensemble machine learning models [8]. The U.S. Drought Monitor literature also motivates our ordinal severity setting: drought categories combine multiple physical indicators and expert judgment, and transitions are usually incremental but can change rapidly under heat and rainfall shocks [9].

II. PROPOSED METHOD (1-3 PAGES)

A. Problem Formulation

Let $\mathcal{R}$ be the set of regions and let $\mathbf{x}_{r,t} \in \mathbb{R}^D$ be the meteorological feature vector for region r on day t . Weekly drought labels are observed only on score days, so most daily rows have missing labels. Given the final test window $\mathcal{T}_r^{test}$ of 91 days for region r , we predict five future weekly scores:

$$\hat{\mathbf{y}}_r = (\hat{y}_{r,1}, \hat{y}_{r,2}, \hat{y}_{r,3}, \hat{y}_{r,4}, \hat{y}_{r,5}). \quad (1)$$

The target metric is

$$\text{MAE} = \frac{1}{5|\mathcal{R}|} \sum_{r \in \mathcal{R}} \sum_{h=1}^5 |\hat{y}_{r,h} - y_{r,h}|. \quad (2)$$

B. Leakage-Aware Feature Construction

The most important engineering constraint is that score labels inside the 91-day blind window are unavailable at inference time. Therefore all features that depend on `score` are computed only from history ending at least 91 days before

the forecast origin. This rule also applies to validation: blind backtests explicitly mask the 91-day window before rebuilding features. Within this contract, the feature table contains:

- **Weather lags:** shifted daily values at 7, 14, 21, 28, 35, 42, and 49 days.
- **Rolling/EWM weather:** rolling mean and standard deviation over 7–35 days and exponentially weighted means with spans 3, 7, and 14.
- **Long drought proxies:** 30/90/180-day precipitation sums, dry-day counts, dry-day share, max dry streak, temperature means, dew-point depression, wet-bulb depression, and related heat/dryness indices.
- **Calendar and seasonality:** month, quarter, day-of-year, week-of-year, and cyclic sine/cosine encodings.
- **Climatology anomalies:** region-month normal weather and standardized deviations from those local seasonal norms.
- **Blind-gap score history:** lagged score summaries shifted by at least 91 days, so the model sees long-term severity context but not hidden test-window labels.

Several early experiments used names containing `two_stage`; these names are retained only for artifact compatibility. The active implementation is a direct-horizon supervised pipeline, not a separate score-reconstruction system.

C. Model Families

For each horizon, we train one independent regression model per family rather than an autoregressive five-week sequence model. For a feature vector $\mathbf{z}_{r,t}$ at forecast origin t , each model family learns an additive tree function

$$F_h(\mathbf{z}_{r,t}) = F_{0,h} + \sum_{m=1}^M \eta_m f_{m,h}(\mathbf{z}_{r,t}), \quad (3)$$

where each $f_{m,h}$ is a regression tree fitted to the residual gradient of the current ensemble. The three libraries share this boosting view, but their tree structure and regularization are different.

LightGBM. LightGBM is our primary tabular anchor. It first quantizes continuous inputs into histogram bins, then searches split gains on those bins instead of on all raw values [2]. Its trees are grown leaf-wise: after every split, the algorithm expands the current leaf with the largest estimated loss reduction. This best-first structure can form deep, specialized rules such as “very low 90-day precipitation in a specific seasonal context,” which is useful for our many lag, rolling, drought-proxy, and climatology features. The risk is overfitting, so the implementation relies on constraints such as learning rate, number of leaves, minimum leaf data, and feature subsampling.

XGBoost. XGBoost also builds an additive ensemble of CART-style regression trees, but the split search is driven by a second-order regularized objective [1]. At each boosting step it approximates the loss with gradients and Hessians, penalizes tree complexity through the number of leaves and leaf weights,

and then applies shrinkage and row/column subsampling. In practice this gives a more conservative, depth-controlled tree structure than LightGBM. It often captures similar nonlinear interactions, but with different split choices, making it a strong diversity partner rather than a duplicate of the LightGBM model.

CatBoost. CatBoost is added because its structure is different in two ways. First, it uses ordered boosting: training examples are processed through random permutations so that target-derived statistics for a row are computed only from earlier rows in that permutation [3]. This is important because `region_id`, `month`, and other calendar buckets are high-cardinality categorical signals that can otherwise memorize labels. Second, CatBoost commonly uses symmetric oblivious trees, where all nodes at the same depth share the same split. These balanced trees are less expressive per tree than arbitrary CART trees, but they are stable and provide a useful categorical-modeling bias for the ensemble.

We used boosted trees deliberately rather than treating them as weak baselines. The course feedback and peer reports suggest that high-capacity neural models can look good under an internal metric but fail to transfer when the public and private distributions differ. Our feature table is heterogeneous, region-heavy, and partly based on hand-audited historical aggregates. Boosted trees let us inspect feature ablations, enforce leakage gaps, and keep exact submission lineage more easily than end-to-end sequence models.

The best reportable submission blends the three model families as

$$\hat{y}_{r,h} = 0.35\hat{y}_{r,h}^{LGB} + 0.35\hat{y}_{r,h}^{XGB} + 0.30\hat{y}_{r,h}^{CAT}. \quad (4)$$

All final predictions are clipped to $[0, 5]$.

D. Validation and Selection Discipline

We used four levels of evidence. First, chronological holdout and rolling-origin validation give fast local feedback. Second, Kaggle-like blind backtests hide a historical 91-day window and score the next five real weekly labels. Third, Kaggle public leaderboard readouts check transfer to the competition test distribution. Fourth, course feedback about public/private gaps was treated as an overfitting warning: a high public score is useful for placement, but it is not sufficient evidence that the modeling method generalizes. These signals often disagreed, so no candidate was promoted from local MAE alone. We also used a leakage sentinel that poisons hidden-window labels and verifies that pseudo-test features do not change. This guard was important because a naive score-persistence model can look excellent locally but fail on Kaggle.

E. Inference and Reproducibility

At inference time, training tail rows and the 91-day test rows are concatenated per region, sorted by synthetic date, and transformed using the saved feature configuration. The last row of the test window becomes the forecast origin for each region. Every candidate submission is sanity-checked for 2,248 rows, exact sample-submission column order, matching

`region_id` order, no NaN values, and prediction range $[0, 5]$. Each important Kaggle score in this report is tied to a CSV path, Kaggle reference id, blend description, and lineage label.

III. EXPERIMENTS (2-4 PAGES)

A. Dataset and Setup

The dataset contains daily meteorological observations over many synthetic years for 2,248 regions. Training has 5,480 days per region and weekly integer drought scores; test has 91 unlabeled days per region. Table II summarizes the data shape.

The raw label table is not a dense daily target. Only 1,757,936 training rows have an observed weekly score, matching the 85.73% score-missing ratio in Table II. Among observed labels, the target is heavily imbalanced: 59.63% are score 0, 17.26% are score 1, and only 1.82% are score 5. This distribution shaped the metric strategy. MAE can be improved by predicting the dominant low-severity class, but the private leaderboard can punish models that miss rare high-severity regions. For this reason, blind backtests also tracked zero-score, nonzero-score, and high-severity slices instead of relying only on the overall average.

The original weather fields also show useful seasonality and distribution shift, shown in Fig. 1. Mean labeled score is lowest around May–June (about 0.75) and highest around August (about 0.94), and the positive-label rate follows the same seasonal pattern. The test weather distribution is visibly warmer than the full training sample and has a large mass at zero precipitation. Therefore, month features alone are insufficient: the model also needs recent rolling weather summaries, precipitation deficit features, heat-dryness interactions, and a validation design that compares the test period to seasonally matched historical windows.

Train-test distribution analysis confirmed that the real test window is not a simple continuation of the global training distribution. Compared with the recent 1,095-day training tail, test is hotter by about 5.35 degrees on average and slightly drier by precipitation and dry-day indicators. Against the stricter same-region, same-month-day previous-year baseline, daily precipitation drops from 3.250 to 1.925, dry-day share rises from 0.403 to 0.487, 90-day precipitation falls from 277.676 to 223.710, and 90-day dry days rise from 35.592 to 43.770. Figure 2 also shows that 90-day rain days have the largest PSI among the plotted variables (0.663), followed by 90-day dry days (0.463). This motivated tail-length, season-matching, and drought-proxy experiments rather than all-history training.

This directly matches the course suggestion to treat seasonality, recent-year importance, and data-range choice as first-class modeling decisions. However, we did not automatically choose the most recent-tail candidate: several recent-weighted runs improved a local or blind metric but failed to transfer to the public leaderboard. The final reportable lineage is therefore the candidate with the best combined evidence, not the candidate with the best single split.

B. Main Model Results

Table I reports the major legal model families and diagnostic controls. Local MAE values come from different validation protocols, so they are used only within the same protocol. Kaggle public MAE is the common external comparison.

The key lesson is that local validation and public transfer are not monotonic. For example, rolling-origin CatBoost and later neural/TCN variants looked strong locally, but public leaderboard transfer was much worse than the historical LGB/XGB/CatBoost anchor. Conversely, the 35/35/30 ensemble made only a modest modeling change but improved the reproducible public score from 0.8232 to 0.8124. Its blind-backtest MAE was 0.4038, worse than the LGBM-only blind anchor at 0.3549, so we treat 35/35/30 as the best reportable public lineage rather than as the pseudo-private validation winner.

C. Feature and Validation Ablations

Feature ablations with LightGBM lean tail1095 were designed as controlled removals, not as a large leaderboard sweep. The baseline used 401 feature columns, train-tail-days 1095, recency half-life 1095, three rolling-origin folds, and refit-full training. Table III and Fig. 3 show the detailed readout.

The largest negative controls were `minus_score_history` and `minus_long_drought_proxy`. Removing either group degraded average rolling MAE by about 0.052. The horizon shape is also informative: `minus_score_history` becomes worse toward weeks 4–5, suggesting that old drought severity labels encode slow regional vulnerability and unobserved geography, while `minus_long_drought_proxy` hurts every horizon because accumulated rain deficit, dry streaks, and heat stress are not redundant with short weather lags.

The most important false positive was `minus_climatology`. It improved rolling-origin MAE from 0.3124 to 0.3046, which would normally look like a clean feature-pruning win. However, its Kaggle-like blind backtest deteriorated to 0.4463, compared with 0.3549 for the baseline blind LGBM anchor. It also had high-severity blind MAE 1.7394. This result changed the selection rule: a feature group could be dropped only if it improved both rolling validation and blind validation, because rolling-origin alone was vulnerable to season-specific overfitting.

`minus_domain_indices` was a smaller but still useful diagnostic. Its average loss was only +0.0020, and early horizons were not clearly worse, but week 5 degraded to 0.3541. We therefore kept the domain indices because the competition averages five future weeks and the late horizons are most exposed to private shift. `minus_region_stats` was recorded as a no-op because that feature switch was disabled in the lean profile; keeping this row in the ablation log prevented us from incorrectly claiming a real experimental effect.

TABLE I
MAIN EXPERIMENT RESULTS AND DECISIONS

Experiment	Feature / model idea	Validation	Local MAE	Public MAE	Decision
Weather-only direct	Direct LGB/XGB without score history	Holdout	0.6770 / 0.7320	0.8640	Rejected as primary method; weather alone misses unobserved regional severity context.
LGB/XGB 50/50 anchor	Score-history boosted-tree ensemble	Holdout	0.6942 / 0.7150	0.8232	First reproducible legal anchor.
CatBoost tail2737	Direct CatBoost with native region/calendar categories	Rolling origin	0.2212	Component only	Useful diversity model; validation scale differs from holdout.
LGB/XGB/Cat 35/35/30	Three-model convex ensemble	Blind backtest + Kaggle sanity	0.4038 blind	0.8124	Best reportable method claim; not blind-best, Kaggle ref 52698259.
Blind LGBM tail1095	Refit-full LGBM, lean features	Blind backtest	0.3549 blind	0.9380	Strong pseudo-private anchor but poor public transfer.
Feature-fused TCN	91-day sequence model with visible time-safe priors	Blind backtest	0.3508 blind	0.9450	Neural diversity signal only; not a public anchor.
Initial accidental hybrid	Leakage-adjacent score behavior	Diagnostic	0.2100	0.8094	Excluded from reportable claims.
Naive score persistence	score gap 0 / forward-fill control	Diagnostic	0.2321	1.0815	Demonstrates severe leakage risk.

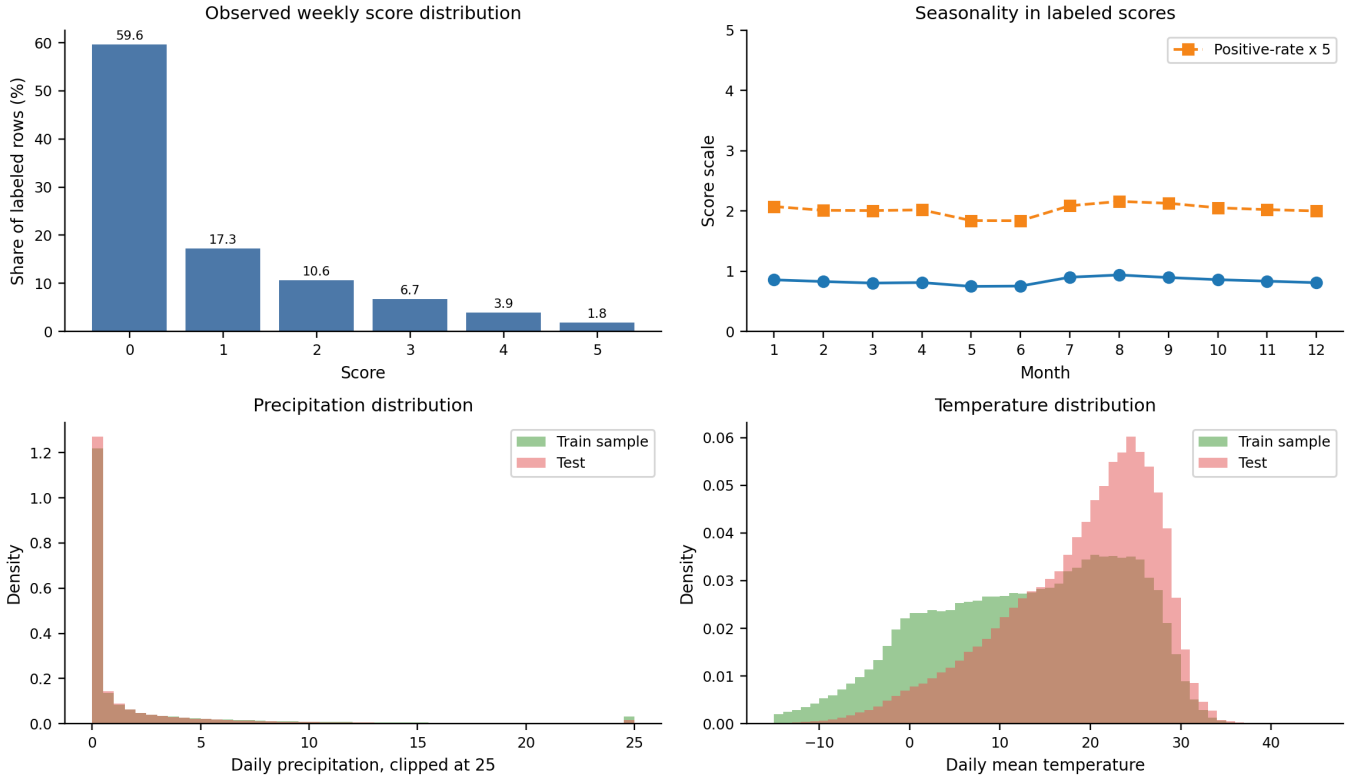


Fig. 1. Exploratory data analysis from raw train/test files. The observed weekly labels are sparse and imbalanced; score 0 accounts for 59.63% of labeled rows, while scores 4 and 5 together account for only 5.77%. The weather panels show that the public test window is warmer and precipitation-light relative to the full training sample.

TABLE II
DATASET STATISTICS

Property	Train	Test
Days per region	5,480	91
Regions	2,248	2,248
Rows	12,319,040	204,568
Raw weather columns	14	14
Score range	0–5	N/A
Score NaN ratio	85.73%	N/A

The ablations also changed our feature discipline. More features were not automatically better: target-history features

were useful but dangerous, so they were allowed only behind the 91-day blind gap; climatology features were kept only when they survived blind checks; and leakage-adjacent artifacts were excluded even when their public score was attractive. This mirrors the peer warning that feature quantity can become a liability without validation-based pruning.

Missingness and shift experiments on 2026-05-23 also failed to replace the reportable anchor. The best validation/blind candidate, the season-matched LGBM shift run, reached rolling MAE 0.24574 and blind MAE 0.38370, but the recent pattern was that validation-strong families transferred poorly to public

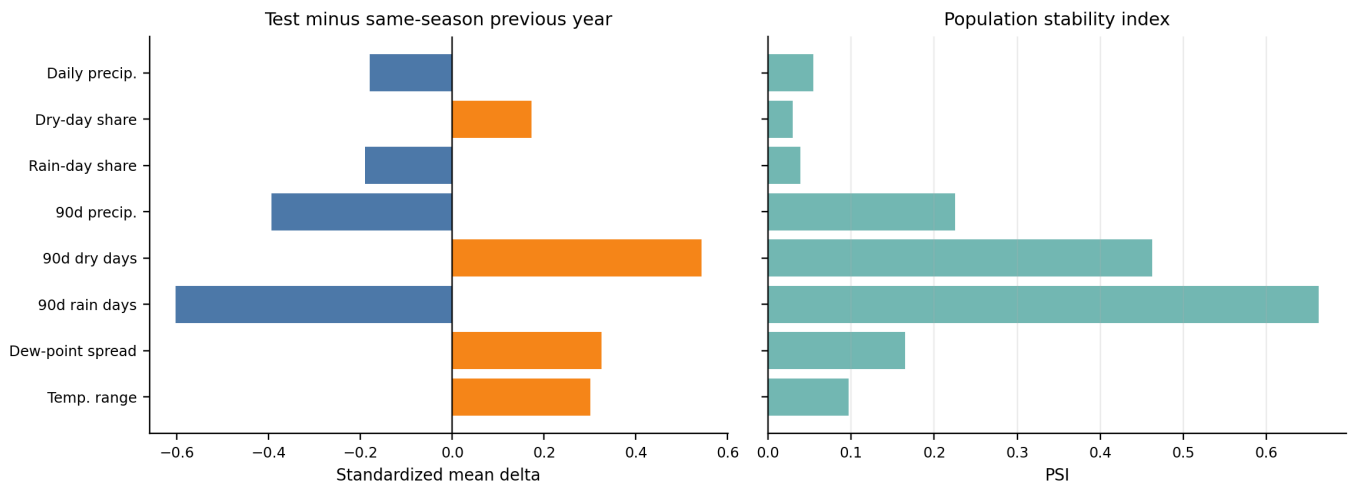


Fig. 2. Train-test shift against the stricter same-region, same-month-day previous-year baseline. The left panel uses standardized mean deltas so features with different units are comparable; the right panel shows PSI. Long-window rain and dry-day features shift more than most daily raw variables, motivating rolling drought proxies and season-matched validation.

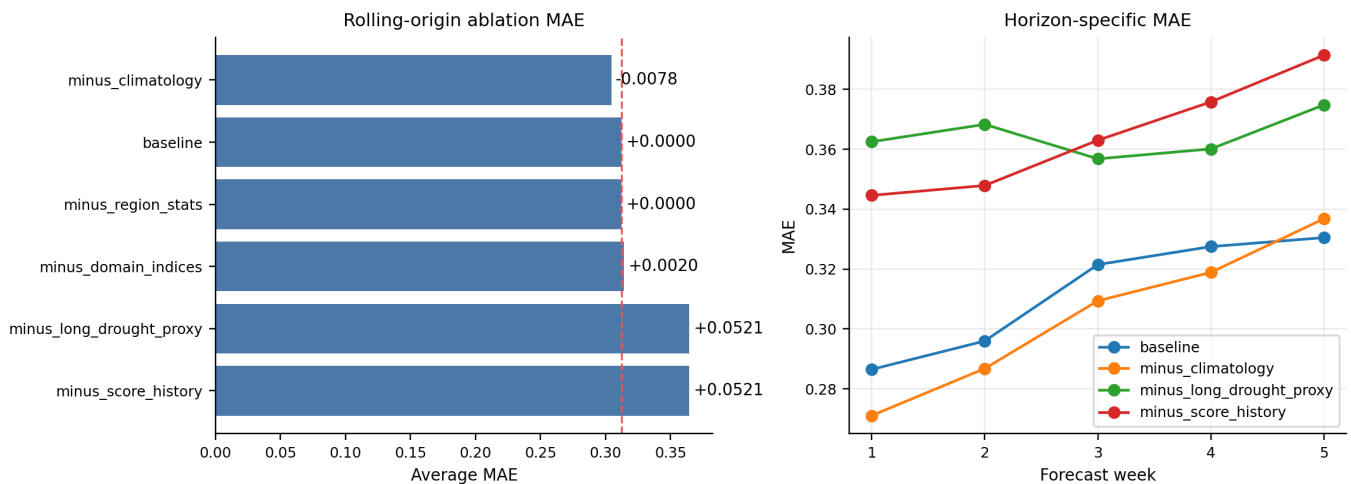


Fig. 3. LightGBM lean tail1095 feature ablation. The left panel compares rolling-origin average MAE against the baseline; the right panel shows how the largest feature removals affect each forecast horizon.

TABLE III
LIGHTGBM LEAN TAIL1095 FEATURE ABLATION DETAILS

Run	Feat.	Avg	Δ	W1	W2	W3	W4/W5	Readout
minus_climatology	386	0.3046	-0.0078	0.2711	0.2868	0.3094	0.3189/0.3368	Local improvement, but blind MAE 0.4463 vs. baseline blind 0.3549; rejected.
baseline	401	0.3124	+0.0000	0.2865	0.2960	0.3215	0.3275/0.3305	Reference configuration.
minus_region_stats	401	0.3124	+0.0000	0.2865	0.2960	0.3215	0.3275/0.3305	No-op because region stats were disabled in this profile.
minus_domain_indices	367	0.3144	+0.0020	0.2794	0.2958	0.3211	0.3217/0.3541	Small average loss; week 5 worsens, so not promoted.
minus_long_drought_proxy	380	0.3645	+0.0521	0.3625	0.3683	0.3568	0.3601/0.3748	Reject; accumulated drought proxies carry strong signal.
minus_score_history	379	0.3645	+0.0521	0.3446	0.3479	0.3630	0.3758/0.3914	Reject; 91-day-gapped score history captures persistent regional state.

LB. We therefore kept it diagnostic rather than promoting it to the main claim.

Table IV summarizes the main knobs that we changed after inspecting either validation failures or live Kaggle feedback.

These were not blind automatic sweeps: each round changed a small number of weights or feature contracts, then we decided whether the new evidence was reportable, diagnostic, or only useful for final selection.

TABLE IV
PARAMETER SENSITIVITY CHECKS

Knob	Values or variants tried	Readout and decision
CatBoost blend weight	10, 20, 30, 35, and 40% CatBoost inside the LGB/XGB/Cat family	Public MAE was 0.8163, 0.8126, 0.8124 , 0.8141, and 0.8168, so 30% became the reportable anchor.
Horizon-specific Boost ramp	Cat- More CatBoost weight on later forecast weeks	Public MAE 0.8138; useful robustness probe, but weaker than 35/35/30.
Feature groups	Dropped climatology, score history, long drought proxies, domain indices, and region stats in LGBM lean tail1095	Score history and long drought proxies were essential; the climatology drop improved rolling MAE but failed blind validation.
Public-chase alphas	horizon Moved exact public-strong history toward the clean 0.8124 anchor, then adjusted early-week and late-week anchor weights	Public MAE improved from 0.8049 to 0.7991, then to 0.7930 and finally 0.7905; label as public-chase, not reportable method evidence.

D. Leaderboard and Final-Selection Readout

Table V shows the relevant public leaderboard snapshot from a live Kaggle CLI re-check on 2026-06-04. The public leaderboard is useful for Kaggle placement, but it can also reward overfitting to public feedback. Therefore we report two separate artifacts: the reportable model lineage and the selected public-chase artifact. A 5/29 static private snapshot also showed Team 5 at rank 5, so the final-selection decision keeps both a public-best artifact and a near-public private-robust hedge.

TABLE V
PUBLIC LEADERBOARD SNAPSHOT

Entry	Public MAE	Rank
Team 10	0.7371	1
Team 2	0.7598	2
Team 3	0.7628	3
Team 15	0.7754	4
Team 4	0.7810	5
Team 20	0.7836	6
Team 14	0.7904	7
Team 5 public-chase	0.7905	8
Team 1	0.7909	9
Team 11	0.8010	10
Team 8	0.8027	11
Baseline 3	0.8056	12
Team 5 reportable lineage	0.8124	–
Baseline 2	0.8623	–
Baseline 1	0.9117	–

The selected public-chase artifact is `submissions/baseline3_private_hedge_v8_cat35_w1_0p0375_keep_shape_horizon_0p0375_0p375_0p55_0p75_1.csv` (Kaggle ref 53204258, SHA-12 390565517cb3, public MAE 0.7905). It is an exact-history horizon hedge moving the recovered 0.8094 public reference toward the clean 0.8124 anchor by horizon alphas $[0.0375, 0.375, 0.55, 0.75, 1.00]$. It is useful for final selection, but it is not a new reportable training algorithm. The current near-public private-risk alternative selected for the second Static Private slot is

`submissions/baseline3_private_hedge_v9_cat35_w1_0p075_w2_0p425_stronger_late_anchor_horizon_0p075_0p425_0p6_0p825_1.csv` (Kaggle ref 53259683, SHA-12 262f683b87b6, public MAE 0.7909). Its horizon alphas are $[0.075, 0.425, 0.60, 0.825, 1.00]$, so it gives up only 0.0004 displayed public MAE while moving week 2 and later horizons closer to the clean anchor. The v8 private hedge ref 53204319 remains the pure-delta fallback, and the previous v7 pair, refs 53186508 and 53186571, remains a fallback if the team wants the earlier manual selection.

Because private leaderboard risk remains possible, the post-crossing workflow did not stop at the public crossing. The v9 follow-up consumed the expanded 10/day quota on 2026-06-01 after all files passed sanity checks; the current manual final-selection recommendation is to select refs 53204258 and 53259683. Ref 53038036 remains a stronger private fallback if a larger public gap is acceptable. These files are sanity-checked final-selection artifacts only, not new reportable model claims.

The reportable artifact remains `submissions/ensemble_20260516_lgb_xgb_cat2737_35_35_30.csv` (Kaggle ref 52698259, SHA-12 bee6f618828d when restored locally). Its prediction distribution is valid and conservative: week means are approximately 0.955, 0.927, 0.880, 0.831, and 0.817, and all predictions are clipped within $[0, 5]$.

E. Discussion

The experiments reveal a “Kaggle paradox.” Pure weather models and some validation-strong models look reasonable locally but underperform on the public leaderboard. Historical score features, when shifted safely beyond the 91-day blind gap, provide information about persistent regional vulnerability that raw weather alone cannot reconstruct. However, target history also creates the largest leakage risk, so the final workflow treats leakage sentinels, blind backtests, and exact artifact lineage as part of model selection rather than as after-the-fact documentation.

Peer reports support the same conclusion from different angles. One group found that discarding older years improved public score but raised public/private overfitting concern; another used LightGBM, TCN, and feature-fused TCN as an ensemble, where LightGBM still added marginal value despite not being the main sequence model; a third group tried GRU, LSTM, CNN, Transformer, and tree models, but reported that neural models did not transfer as reliably as their internal metrics suggested. Our response is therefore not to maximize model complexity, but to emphasize distribution checks, leakage-safe features, heterogeneous tree ensembles, and conservative artifact labeling.

The public-chase sequence further shows why leaderboard scores must be labeled carefully. It improved Team 5 from the reportable 0.8124 anchor to 0.7905 by using public feedback around exact historical artifacts. This is acceptable for Kaggle final selection if the competition UI allows selecting the

submitted artifact, but it should not be described as evidence that the supervised training method itself reached 0.7905.

In plain language, our final interpretation is that the strongest transferable modeling idea is still the leakage-safe boosted-tree ensemble, while the best Kaggle score comes from small, manually reviewed horizon-weight adjustments around known submitted shapes. That is why the report keeps the 0.7905 and 0.8124 numbers side by side instead of merging them into one claim.

IV. CONCLUSION

We built a leakage-aware drought severity forecasting pipeline for the Data Mining final project. The final reportable method uses direct-horizon LightGBM, XGBoost, and CatBoost models over carefully shifted temporal, meteorological, climatology, and historical severity features. The best reproducible modeling claim is the 35/35/30 boosted-tree ensemble with public MAE 0.8124. Extensive diagnostics showed that score history and long drought proxies are essential, that local rolling-origin MAE can be misleading under train-test shift, and that blind-window leakage controls are necessary. A separate public-chase artifact reaches public MAE 0.7905 and crosses Baseline 3; we keep it available for final leaderboard selection while preserving the clean 0.8124 lineage as the scientific method claim.

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